Vanishing Point

Project Overview: Phase 1 (Simulator and Seeker), Phase 2 (PPO Evader) and Stage 2 (learning both drones)

1. Project in one paragraph

Vanishing Point trains an Evader drone to stay out of a Seeker drone's camera view in a bounded 3D arena (20 x 20 x 10 m, 0.1 s timestep, 600-step episodes). Phase 1 built the simulator, a scripted Seeker, and a random-Evader baseline. Phase 2 trained the Evader with PPO, first plainly and then with a difficulty curriculum, and validated the results across multiple seeds. Stage 2 then replaced the scripted Seeker with a learning Seeker: both drones are trained with model-free, policy-gradient actor-critic methods (MADDPG, with MAPPO as a comparison) in a world that now includes solid obstacles that block line of sight. Stage 2 is complete in simulation, trained and evaluated on 5 independent seeds with held-out opponents; the physical-drone phase of the original proposal is not done and is documented as future work (Section 7).

> Rules the project follows: no value iteration, no policy iteration and no planning formulation; policy-gradient, model-free learning only. The scripted Seeker belongs to Stage 1; **no part of Stage 2 training or evaluation uses it**.

2. Phase 1: building the simulator and the scripted Seeker

2.1 First code version

The first simulator version was a clean single file with correct FOV geometry and simple velocity-controlled drones, but it had several problems that made its results unreliable: a wall penalty of -10 per step that dwarfed the +/-1 visibility reward, a fixed random seed (7) that made every run identical, and a Seeker that simply gave up and span in place after losing the Evader, with a scan pattern that left a blind spot above and below.

Metric	Value (seed 3167782036)
Total reward	426.0
Undetected time	86.2%
Time to first detection	0.0 s (uninformative metric)
Wall touches	4

With this version, the Seeker locked onto the Evader early (by step 80), lost it around step 100 when the Evader drifted 64 degrees off-angle, and then never found it again for the rest of the episode, about 517 of 600 steps hidden. That is a misleading baseline: the high undetected percentage reflects a Seeker that gave up, not an Evader that is hard to catch.

2.2 Fixing the Seeker

The Evader was left random and the Seeker was tuned into a fair opponent before any PPO training began:

Change	Before	After	Why
Wall penalty	-10 every step against a wall	-2 once on first contact	Keeps hiding/being-seen as the main signal
Random seed	Fixed at 7	New each run, logged	Shows real variation, allows replay
Seeker modes	Follow, then spin forever	TRACK, SCAN (4 s timeout), SEARCH	Recovers after losing the Evader
Scan pattern	Left/right only	Spin + 45 degree tilt	Removes blind spot above/below
Start behaviour	Slow forward patrol	Scan first, then search	Avoids getting stuck at a wall

Change	Before	After	Why
Metrics	Time to first detection	Time until first loss, longest hidden stretch, wall touches, search restarts	Describes how an episode unfolds

Note added after code review: the wall penalty fires at the start of each wall-contact streak, so leaving a wall and touching it again fires it again. It is not charged for every step spent against a wall, which later turned out to matter (Section 4.5).

2.3 Results after tuning

Metric	First version	Tuned version (seed 3171674840)
Total reward	426.0	-256.0
Undetected time	86.2%	30.3%
Time until first loss of sight	~8 s (inferred)	8.6 s
Longest undetected stretch	~50 s (inferred)	8.3 s
Wall touches	4	10
Seeker search restarts	n/a	1

The reward sign flipped because the fixed Seeker now sees the random Evader about 70% of the time, giving PPO a real benchmark to beat instead of a Seeker that could be escaped by accident. A 20-seed headless test confirmed the improvement generalises rather than being a one-off:

Metric (20-seed average)	Old Seeker	New Seeker
Undetected time	56.0%	37.7%
Longest hidden stretch	22.2 s	13.4 s
Worst single stretch	50.0 s	48.9 s

The worst single hidden stretch barely moved (50.0 s to 48.9 s), showing the Seeker's random search waypoints still leave it blind on unlucky seeds, a known weakness carried into Phase 2.

2.4 Phase 1 pros and cons

- **Pros:** recovers after losing the Evader via SCAN/SEARCH, scans the full 3D space, reproducible seeds, a meaningful opponent and baseline for PPO.
- **Cons:** random search waypoints still leave long worst-case hidden stretches; the Seeker (1.8 m/s) is slower than the Evader (2.2 m/s) and does not predict movement; drones can overlap with no penalty.

3. Phase 2: training the Evader with PPO

Three controllers were compared against the same tuned scripted Seeker: acting randomly, PPO trained the plain way for 500,000 steps, and PPO trained the same length with a curriculum that starts against a weak Seeker and strengthens it across four stages. Each setup was run across 3 training seeds (1 original + 2 more added later) and evaluated on the same 20 episodes at full difficulty.

3.1 Results per seed

Setup	Seed	Mean return	Undetected	Wall touches
Random baseline	12345	173.3	66.0%	9.25
Plain PPO	577448602 (original)	466.5	89.6%	4.20
Plain PPO	2001	470.5	90.0%	4.55
Plain PPO	2002	472.7	91.0%	9.50
Curriculum PPO	577448602 (original)	458.8	88.7%	2.60
Curriculum PPO	2001	452.6	88.2%	3.00
Curriculum PPO	2002	447.0	87.9%	3.80

3.2 Averaged comparison

Metric	Random baseline	Plain PPO (avg of 3 seeds)	Curriculum PPO (avg of 3 seeds)
Mean return	173.3	469.9	452.8
Undetected time	66.0%	90.2%	88.3%
Time until first loss	3.5 s	1.7 s	1.8 s
Longest hidden stretch	10.7 s	33.3 s	27.4 s
Wall touches (mean)	9.25	6.08	3.13
Seeker search restarts	1.30	5.55	4.78

PPO is a large jump over random behaviour on every metric: undetected time rises from 66.0% to about 90%, sight is lost almost immediately (1.7-1.8 s versus 3.5 s), and wall touches fall. Between the two PPO variants, the pattern held across all three seeds without exception: plain PPO scored higher every time, and curriculum PPO touched walls less often every time. Averaged out, plain PPO wins on return (469.9 vs 452.8) and undetected time (90.2% vs 88.3%), while curriculum PPO roughly halves wall touches (6.08 vs 3.13). This is a repeatable trade-off rather than single-seed noise.

A plausible explanation (no ablation was run to confirm it). Reward only measures undetected time, and the wall penalty fires once per contact rather than scaling with how close or how often the drone grazes a wall, so a policy that clips corners to break line of sight a step or two earlier is not punished much for the risk it takes. Training the whole run at full Seeker speed and field of view may let plain PPO find and keep that riskier shortcut, while curriculum training spends its early stages against a slow, narrow-FOV Seeker, where survival depends on clean navigation. The training logs are consistent with this: under curriculum, ep_rew_mean recovers within a few thousand timesteps after each difficulty increase, and wall-touch variance across seeds stays in a tight band (2.60-3.80) while plain PPO ranges far wider (4.20-9.50), including one seed (2002) that settled on a distinctly wall-heavy strategy.

3.3 Reward convergence

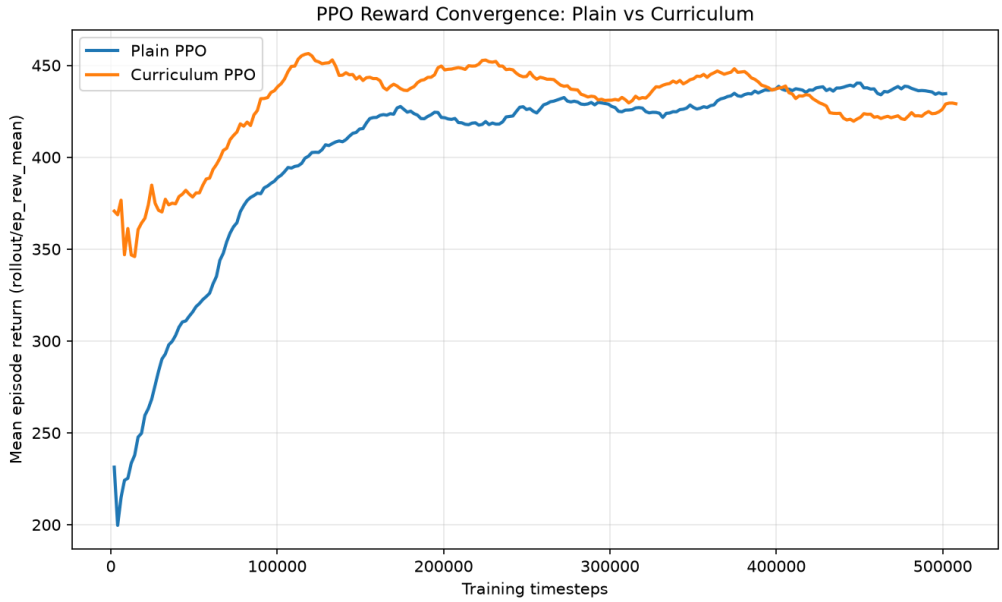


Figure 1. Stage 1 (open arena) reward convergence, plain vs curriculum PPO, one seed each.

Curriculum training starts far higher since its early stages face an easy, slow, narrow-sighted Seeker, so an early lead there is expected, not meaningful. Both curves reach a similar plateau by the end, and curriculum PPO dips slightly below plain PPO in the final stretch, which lines up with curriculum stage 4 (full difficulty) arriving late.

3.4 Independent tests

Automated tests were added to check behaviour that training runs alone could not verify: the FOV boundary, Seeker mode transitions, the wall penalty, valid resets, correct episode length, and the absence of NaN/infinite values. All eleven Stage 1 test cases still pass; the full project suite is now 74 tests (Section 4.9).

Test area	Result
FOV boundary (visible/not visible at edges and range)	Pass
Seeker mode transitions (visible to TRACK, timeout to SEARCH)	Pass
Wall penalty fires once per contact streak, not once per step in contact	Pass
Reset returns a valid, bounded observation	Pass
Episode truncates at exactly 600 steps	Pass
No NaN or infinite values across random episodes	Pass

3.5 Phase 2 pros and cons

- **Pros:** PPO massively outperforms random behaviour; the Evader loses the Seeker almost immediately and touches walls far less; curriculum training is confirmed (3 seeds) to trade a small amount of score for fewer wall collisions.
- **Cons:** the Seeker is still slower than the Evader, so part of the score may reflect out-running rather than pure evasive skill; the Evader's observation excludes the Seeker's current mode; only 3 seeds were tested, and only at full difficulty for the final comparison.

4. Stage 2: learning both drones

4.1 Goal and design rules

Stage 2 asks whether the Seeker can learn its own search behaviour while the Evader learns to evade it. Both agents are neural policies trained model-free with actor-critic policy-gradient updates. Nothing in Stage 2 uses a scripted controller, a model of the environment, value/policy iteration or planning. The only fixed opponents during training are a frozen PPO Evader (also a policy-gradient method) and a frozen, previously trained learned Seeker.

4.2 The Stage 2 environment

- **Shared simulator.** One NumPy world drives both the Gymnasium (PPO) and PettingZoo (multi-agent) interfaces. A regression check against the original first-commit code (random and fixed starts, three difficulty levels, 10,800 transitions) found a maximum difference of exactly 0, so Stage 1 behaviour is preserved.
- **Actions.** Evader: 3D velocity (capped at 2.2 m/s). Seeker: 3D velocity (capped at 1.8 m/s) plus yaw and pitch rates (100 and 90 degrees per second, pitch limited to 89 degrees). Unlike the scripted Seeker, a learned Seeker's camera turns at a limited rate.
- **Observations.** Evader: 21 values (as in Stage 1, so saved PPO models stay compatible). Seeker: 21 values with only its own state, its camera, and the last-seen position of the Evader, never the Evader's live position when hidden. The Seeker also receives 4 past observations spaced 5 steps apart (the observation history).
- **Centralised critics (training only).** Each agent's critic sees the 26-value global state and both agents' actions; execution uses only local observations.
- **Rewards.** Evader +1 hidden, -1 visible; the Seeker gets the opposite. Wall contact costs -2 at the start of a contact streak. For training, an extra per-step penalty (2.0) is charged while a drone stays in contact with a wall or obstacle; every evaluation uses the original unshaped rewards.

4.3 The obstacle world

To make the task less plain, the arena can contain solid boxes (preset *city*: a 6 m central wall, two full-height pillars, two low blocks and a low canopy). Obstacles block movement (collisions are scored like wall contacts) and block line of sight: a drone inside the view cone and range is still hidden if a box lies between the two drones. Each actor also receives the vectors to its three nearest obstacles (9 extra values). With obstacles off, every Stage 1 behaviour is unchanged.

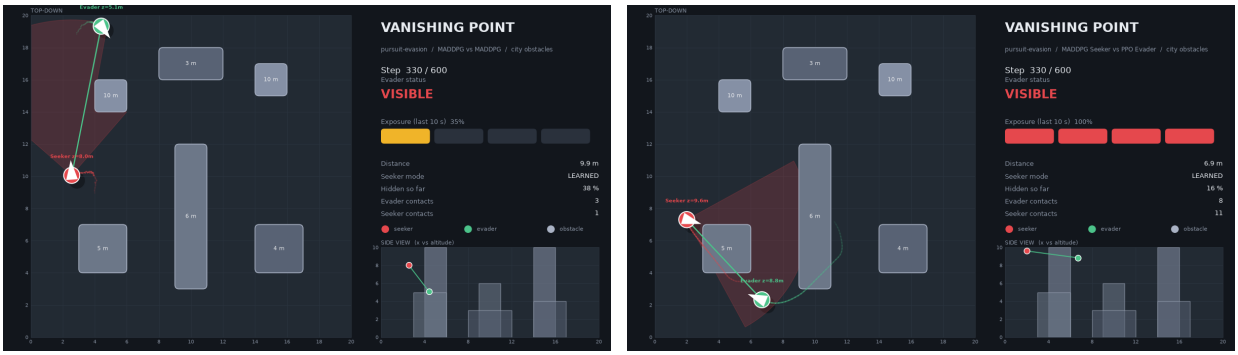


Figure 2. The 2D viewer on the obstacle world (top-down map, side elevation, statistics sidebar). Left: the two jointly trained agents. Right: the Seeker against the held-out PPO Evader. The shaded wedge is the Seeker's view cone, cut off behind obstacles; the line shows a visible (green) sight line.

4.4 Method and final training recipe

MADDPG (Lowe et al., 2017): one actor per agent (two 128-unit layers, tanh), one centralised critic per agent, target networks with soft updates (tau 0.01), a replay buffer of 100,000 transitions, Gaussian exploration noise decaying from 0.3 to 0.05, discount 0.99. Time-limit endings bootstrap from the final state instead of the reset state.

Component	Setting in the final recipe
Phase B (Seeker pretraining)	Seeker alone against a frozen PPO Evader, 400k steps
Phase C (joint training)	600k steps, warm start from Phase B, one learner at a time in blocks of 20k steps
Opponent mixing (per episode)	50% frozen PPO Evader, 25% frozen Phase B Seeker, 25% joint game
Seeker observation history	4 past observations, 5 steps apart
Wall/obstacle penalty (training only)	2.0 per step in contact
Action regulariser (training only)	0.05 on actor pre-tanh outputs, prevents saturated actions
Evaluation opponents	PPO Evader of a different seed (2002) and learned Seekers of other training runs

4.5 What went wrong first, and what fixed it

The original Stage 2 pipeline ran but produced a degenerate result, which is worth recording because it shaped the final design. In joint training the Seeker parked in a corner with its camera pointing at the floor, and both drones sat against walls for about 500 of 600 steps. The reward function explains it: the wall penalty only fires once per contact, so once a drone touches a wall it can stay there for free. (The Stage 1 PPO Evader also spends about 126 of 600 steps on walls, so the behaviour was latent from the start.)

Change	Effect (development runs, one seed, 200k steps)
Per-step wall penalty at least as large as the +1 hidden reward	Removes corner parking. 0.5 was too small and collapsed on another seed; 3 and 5 hurt the Seeker
Seeker observation history (4 x 5 steps)	Largest single gain: Seeker vs PPO 45% to 83% visible (3 seeds, dedicated training)
Opponent mixing and Seeker pretraining	Stopped joint training from erasing the Seeker's skill; essential (Section 4.8)
More steps, lower learning rates, more updates per step, regulariser or warm start alone	No reliable improvement
Fix: frozen opponent's next action may enter the OTHER agent's critic target, never its own	A mistake of this kind made the Evader worse until corrected

Two evaluation problems were also found and fixed in a review: training seeds overlapped across runs (episode seeds now come from separate blocks per seed), and the old evaluation reported shaped rewards and a wrong seed count (it now scores unshaped rewards, reports 95% bootstrap confidence intervals, and uses paired starts).

4.6 Baselines in the obstacle world

PPO was retrained inside the obstacle world (it needs the 9 extra observation values), 3 seeds each for plain and curriculum, and evaluated on 50 episodes against the scripted Seeker. The PPO Evader hides about 88% of the time, the same quality as in the open arena.

Setup	Seed 2001	Seed 2002	Seed 2003	Mean
Random (return / undetected)	-	-	-	185.5 / 67.4%
Plain PPO return	444.0	439.5	447.7	443.7
Plain PPO undetected	88.3%	87.7%	88.3%	88.1%

Setup	Seed 2001	Seed 2002	Seed 2003	Mean
Plain PPO wall/obstacle touches	8.00	6.16	5.88	6.68
Curriculum PPO return	443.8	444.0	452.0	446.6
Curriculum PPO undetected	88.3%	87.9%	88.3%	88.2%
Curriculum PPO wall/obstacle touches	8.18	5.42	3.98	5.86

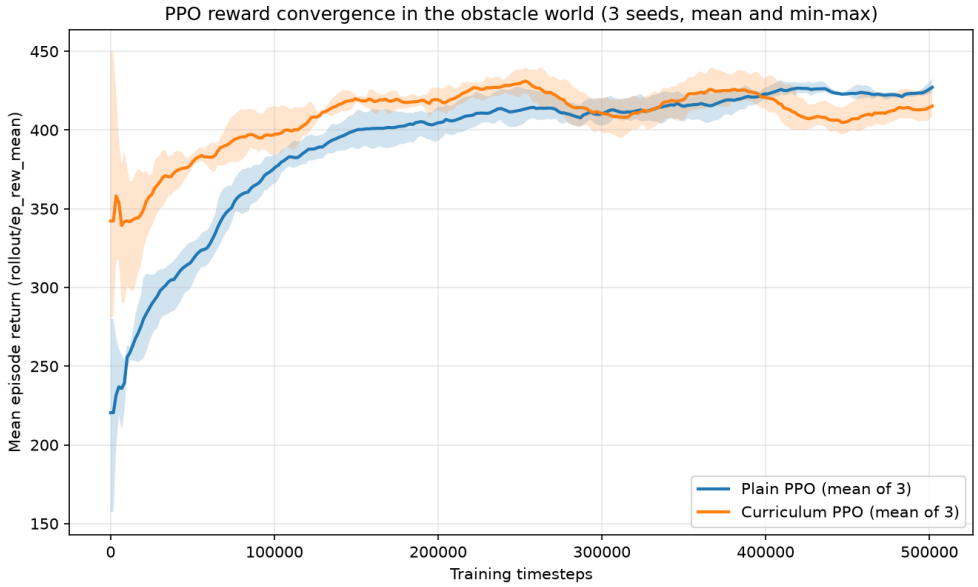


Figure 3. PPO reward convergence in the obstacle world (3 seeds, mean and min-max band). Both variants plateau by about 250k steps, so the extra steps up to 500k add almost nothing.

4.7 Stage 2 results

All numbers are the percentage of the 60 s episode in which the Evader is visible to the Seeker (lower is better for the Evader), on 60 paired held-out starts per matchup per training seed, with original unshaped rewards. The scripted-Seeker reference (Stage 1) is 11.4% [95% CI 10.4, 12.4] visible against the same PPO Evader.

Matchup	Mean	Seed 2001	2002	2003	2004	2005
Phase B Seeker vs held-out PPO Evader	64.1	96.4	17.9	67.6	87.1	51.7
Joint-trained Seeker vs held-out PPO Evader	52.8	80.8	34.7	58.1	11.8	78.7
Joint Seeker vs joint Evader (co-trained pair)	20.2	33.3	7.3	11.6	20.2	28.8
Joint Evader vs a held-out learned Seeker	19.6	5.2	4.9	19.5	41.7	26.7
PPO Evader vs the same held-out Seekers	64.1	17.9	67.6	87.1	51.7	96.4

- A learned Seeker keeps a PPO Evader in view 53-64% of the time on average against 11% for the scripted Seeker, but the spread between seeds is large (12% to 96%).
- The jointly trained Evader is seen far less (19.6%) than the PPO Evader (64.1%) by the same held-out learned Seekers in all five pairings. This comparison is tilted against PPO, because those Seekers were trained against PPO-style Evaders.
- In the co-trained game neither side dominates consistently (7% to 33% visible).

Proposal metrics: time to first detection and violations

Time to first detection is 0 s by definition when the Seeker already sees the Evader at the start, so a second evaluation points the camera away at step 0 (hidden start). Means over 5 seeds x 60 starts = 300 episodes per row:

Matchup	Visible %, visible start	Visible %, hidden start	First detection, hidden start (mean / median, episodes detected)
Phase B Seeker vs PPO Evader	64.1	60.0	4.2 s / 2.0 s (290 of 300)
Joint Seeker vs PPO Evader	52.8	47.1	6.8 s / 2.1 s (280 of 300)
Joint Seeker vs joint Evader	20.2	18.4	6.1 s / 2.2 s (265 of 300)
Held-out learned Seeker vs joint Evader	19.6	14.6	6.2 s / 2.0 s (254 of 300)

Episodes where the Seeker never finds the Evader within 60 s are counted separately and not averaged in: against the joint Evader that is 12-15% of hidden-start episodes.

Violations per 60 s episode (visible start)	Evader contacts	Seeker contacts	Evader steps in contact	Seeker steps in contact	Drone-drone collision steps
Phase B Seeker vs PPO Evader	10.3	8.4	170	68	2.4
Joint Seeker vs PPO Evader	13.0	8.5	196	125	1.8
Joint Seeker vs joint Evader	13.5	14.8	262	155	0.2
Held-out learned Seeker vs joint Evader	13.9	19.1	293	127	0.2

Contacts count new contact streaks with a wall or obstacle; steps in contact count every step touching. Drone-drone collisions (centres closer than 0.7 m) are measured only, not penalised in training; they are rare.

Reward convergence and stability

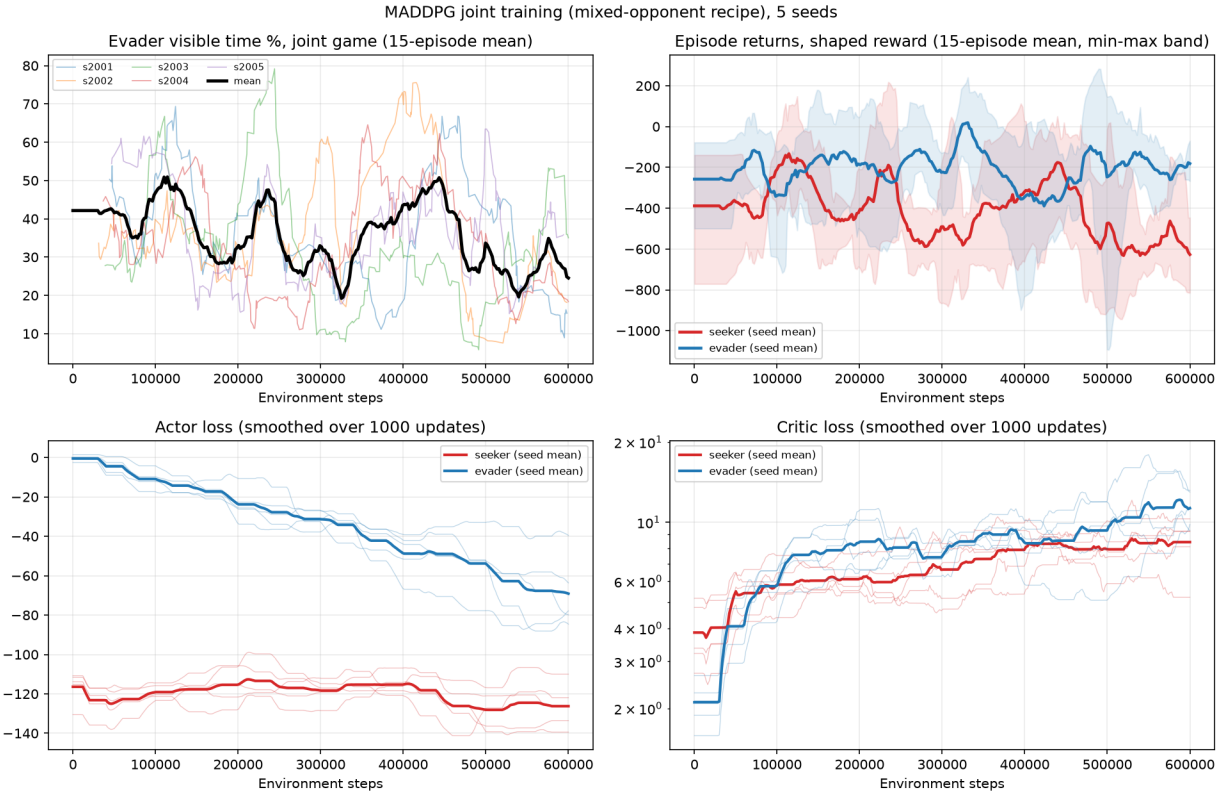


Figure 4. MADDPG joint training, 5 seeds (thin lines) and seed mean (bold). Competing learners keep adapting to each other, so visible time oscillates (individual seeds swing between roughly 5% and 80%) instead of converging; the seed mean ends around 25%.

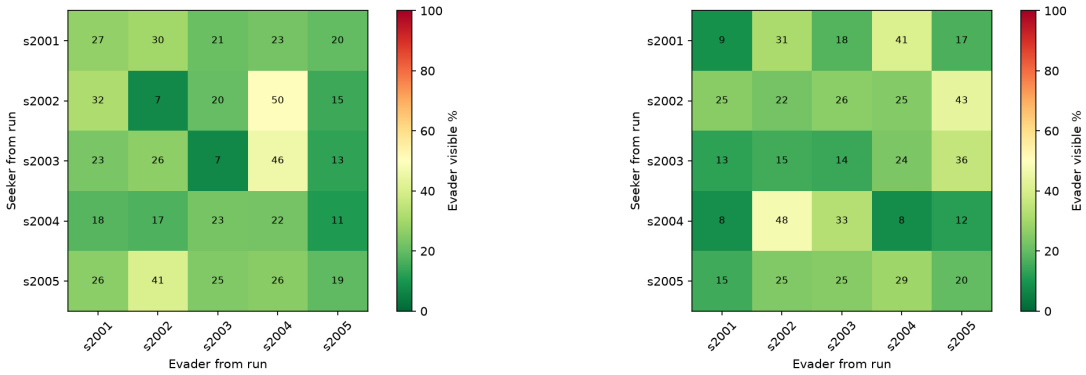


Figure 5. Cross-play: Evader visible % when every Seeker (rows) meets every Evader (columns) from independent training runs. The diagonal is each run's own co-trained pair. Left: main recipe. Right: with an opponent pool (Section 4.8).

4.8 Comparisons: ablations, MAPPO and an opponent pool

These use 3 training seeds (pool: 5), the same budgets, and the same held-out opponents. With n = 3 the seed-to-seed standard deviation (5-40 points) is as large as many differences, so only large effects (more than about 20 points) are clearly real.

Ablations remove one component from the final recipe. Columns: Seeker vs held-out PPO (higher is a stronger Seeker), Evader vs held-out learned Seekers (lower is a stronger Evader), joint game, and steps in contact with walls/obstacles.

Variant	Seeker vs PPO	Evader vs held-out	Joint game	Contact steps (Evader / Seeker)
Final recipe	57.9 +/- 18.8	25.4 +/- 9.0	17.4 +/- 11.4	222 / 167
No Seeker observation history	33.0 +/- 16.8	24.7 +/- 9.0	35.7 +/- 15.8	214 / 131
No wall/obstacle penalty	50.6 +/- 22.8	18.5 +/- 3.6	12.3 +/- 5.5	564 / 408
No opponent mixing	24.9 +/- 6.5	34.4 +/- 14.7	20.3 +/- 13.2	150 / 150
No alternating learner blocks	47.7 +/- 19.7	23.5 +/- 5.0	24.6 +/- 11.5	203 / 187
No Seeker pretraining (cold start)	28.4 +/- 15.1	37.2 +/- 9.6	13.2 +/- 3.4	130 / 136

- **Clearly matter:** opponent mixing, Seeker pretraining and the Seeker's observation history.
- **Wall penalty:** removing it does not hurt the visibility score, but wall and obstacle contact more than doubles (564 / 408 steps): both drones park against walls. Its job is to stop degenerate wall-hugging.
- **Alternating blocks:** the difference is inside the noise.

MAPPO. A comparison method (on-policy clipped PPO, per-agent Gaussian actors, a centralised value critic per agent, GAE) was implemented on the same environment, opponents and budgets.

	Seeker vs PPO	Evader vs held-out	Joint game	Contact steps (Evader / Seeker)
MADDPG (3 seeds)	57.9 +/- 18.8	25.4 +/- 9.0	17.4 +/- 11.4	222 / 167
MAPPO (same 3 seeds)	36.5 +/- 41.1	30.5 +/- 11.0	22.4 +/- 21.2	36 / 33

MAPPO's Seekers overfit the one PPO Evader they trained against: they see it 96%, 95% and 72% of the time, but against a different PPO Evader only seed 2002 holds up (94%); the other two fall to 6-8%. Stochastic and deterministic actions give the same numbers, so this is real over-specialisation, not an evaluation artefact. MAPPO's drones almost never touch walls (about 35 steps versus 170-220 for MADDPG). On this task MAPPO was not better overall: weaker and far more variable Seeker skill, slightly weaker Evader, much cleaner movement.

Opponent pool and stability. Every 30k steps a snapshot of both learners is stored (20 kept); 15% of episodes play the learning Evader against a sampled past Seeker and 10% the learning Seeker against a sampled past Evader (5 seeds).

Measure	Main recipe	With pool
Training oscillation: std of smoothed visible % over the second half (points)	14.7	10.8
Joint-game visible %: mean (std across seeds)	20.2 (9.9)	16.9 (5.4)
Joint Seeker vs held-out PPO: mean (std across seeds)	52.8 (26.4)	49.8 (20.3)
Cross-play, strangers (off-diagonal) mean	25.2	25.4
Cross-play gap: strangers minus own pair	8.6	10.7

The pool gives smoother training and about half the seed-to-seed spread in the joint game, but no gain in average strength and no gain in cross-play robustness. With 5 seeds none of these differences is statistically solid.

4.9 Tests and validation

The suite now has 74 tests in the tests/ folder. Besides the 11 Stage 1 tests: Stage 2 environment (joint dynamics, speed caps, FOV geometry, camera updates, hidden-target non-leakage, wall rewards, truncation, the PettingZoo contract); MADDPG (learning, replay alignment, targets, frozen-opponent and opponent-pool handling, checkpoints, reproducibility); obstacles (line of sight, collisions, sliding, observation sizes, start states, scripted-Seeker freezing, observation history, first-detection time, drone collisions); MAPPO (GAE, masking, checkpoints, training smoke runs); evaluation (paired starts, confidence intervals, seed counts, cross-play). A viewer bug (the status label showing a tuple when the Evader was visible) was found while preparing this report and fixed.

4.10 Stage 2 pros and cons

- **Pros:** rule-compliant (policy-gradient, model-free, no scripted controller); multi-seed with held-out opponents, confidence intervals, ablations and a MAPPO comparison; a learned Seeker clearly beats the scripted one against PPO; every proposal metric is measured.
- **Cons:** large seed-to-seed variance (a Seeker anywhere from 12% to 96% against PPO); co-training oscillates rather than converges; the game is asymmetric (the Evader sees the Seeker's true state and is faster, 2.2 versus 1.8 m/s); the learned Seeker's camera is rate-limited while the scripted one turns instantly, which confounds Stage 1 versus Stage 2 comparisons; wall and obstacle contact stays common for MADDPG drones; most ablations use only 3 seeds; the simulator is point-mass.

5. Which model to use, and current status

For the open arena, plain PPO (seed 577448602, saved as models/ppo_500k) remains the headline Stage 1 result. In the obstacle world the reference opponents are the PPO models models/ppo_city_plain_s2001 (used in training) and s2002 (held out for evaluation). For Stage 2 the recommended system is the final MADDPG recipe (Section 4.4; configs/maddpg_config_final_phaseB.json then configs/maddpg_config_final.json), reported as a mean with its seed spread rather than as one best seed. Seed 2001 is used for the figures in this report. More training steps are not expected to help: PPO plateaus by about 250k steps and the joint MADDPG curves oscillate without trend. Phases 1, 2 and 3 (simulator, PPO with curriculum, simultaneous MADDPG learning) are complete; Phase 4 (physical drones) is not.

6. Alignment with the project proposal

Proposal item	Status	Where
Lightweight arena, point-mass drones, geometric FOV cone	Done	Sections 2, 4.2
Evader trained with PPO against a rule-based Seeker, with a curriculum	Done (3 seeds, open arena and obstacle world)	Sections 3, 4.6. The curriculum raises Seeker speed and widens the FOV from 60 to 90 degrees; the proposal text says it narrows it
Both drones trained simultaneously (MADDPG), repeated runs	Done: 5 seeds, held-out opponents. Implemented as co-training with opponent mixing	Sections 4.4-4.8
Proportion of time undetected	Done	Section 4.7
Average time to first detection	Done with a hidden-start evaluation	Section 4.7
Boundary and collision violation rate	Done: walls, obstacles and drone-drone collisions (the last measured, not penalised)	Section 4.7
Reward convergence over training	Done for PPO and MADDPG; MADDPG oscillates	Figures 3, 4
Deployment on two physical drones	Not done: future work	Section 7
Beyond the proposal	Obstacle world, MAPPO comparison, opponent pool, cross-play, ablations, viewers	Sections 4.3, 4.8

7. Future work: physical drones and next steps

Phase 4 of the proposal (hardware) was not attempted. The proposal makes it conditional on reliable simulation behaviour, and the seed-to-seed variance above means the policies are not yet reliable enough; it also needs two drones, a tracking system and a netted arena. No hardware result is claimed anywhere in this project.

Simulator assumption	Real-drone difference
Point-mass velocity commands applied instantly	Quadrotor attitude dynamics, motor limits, a velocity controller with lag
0.1 s step, no sensing or actuation delay	Latency from tracking, inference and radio
The Evader sees the Seeker's exact state	Needs motion capture or onboard estimation, both noisy
Visibility is a perfect geometric test	A real camera and detector: misses, false positives, motion blur, lighting
Free yaw and pitch rates for the camera	Gimbal or body-yaw limits and lag

Proposed plan: (1) wrap both policies in a velocity-command interface and keep the observation definitions; (2) identify the drone's velocity response and latency and add them to the simulator; (3) retrain with randomised lag, delay, noise and detection misses and re-run the same evaluation; (4) add safety (a geofence smaller than the arena, a hard minimum separation, speed caps, a kill switch); (5) test in stages (one drone against a replayed Seeker camera, then the learned Seeker against a human-flown Evader, then both learned policies); (6) report the sim-to-real gap honestly. Other next steps: recurrent policies instead of fixed observation history, symmetric information, a larger opponent pool, and more seeds (for example 10) to firm up the ablations.

8. Appendix: how to reproduce

All commands assume the project's Python environment (pip install -r requirements.txt; on macOS pybullet may need `CFLAGS="-Dfdopen=fdopen"`) and that the terminal is in the project folder. Outputs go to runs/ (git-ignored).

8.1 Stage 1 (open arena)

```
python -m vp.ppo random --episodes 20
python -m vp.ppo train --timesteps 500000 --seed 577448602 \
    --model models/ppo_500k --tensorboard-log logs/
python -m vp.ppo train --curriculum --timesteps 500000 --seed 577448602 \
    --model models/ppo_curr --tensorboard-log logs/
python -m vp.ppo eval --model models/ppo_500k --episodes 20
python -m scripts.plot_convergence --plain logs/PPO_1 --curriculum logs/PPO_3 --out docs/figures/convergence.png
pytest -q
```

Give every training run its own --tensorboard-log folder if runs are started at the same time; otherwise they race for the PPO_N folder names and interleave their logs.

8.2 Stage 2 (obstacle world)

```
# opponents inside the obstacle world
python -m vp.ppo train --obstacles city --timesteps 500000 --seed 2001 \
    --model models/ppo_city_plain_s2001 --tensorboard-log logs_city/curve_plain_2001
# Phase B: Seeker alone against a frozen PPO Evader
python -m scripts.train_maddpg --phase seeker --config configs/maddpg_config_final_phaseB.json \
    --opponent ppo --opponent-model models/ppo_city_plain_s2001.zip \
    --output runs/B --seeds 2001
# Phase C: joint training with opponent mixing
python -m scripts.train_maddpg --phase simultaneous --config configs/maddpg_config_final.json \
    --initialize runs/B/seeker_s2001/final.pt --output runs/C --seeds 2001
# evaluation: 60 paired starts, unshaped rewards, bootstrap CIs
# (add --hidden-start to measure time to first detection)
python -m scripts.evaluate_stage2 --ppo-model models/ppo_city_plain_s2002.zip \
    --seeker-checkpoint runs/B/seeker_s2001/final.pt \
    --joint-checkpoint runs/C/simultaneous_s2001/final.pt --output runs/eval
# stability, ablations and the MAPPO comparison
python -m scripts.crossplay_stage2 --checkpoints <runs...> --labels <names...> --output runs/cross
python -m scripts.ablation_report A0_final A1_nohist ...     # reads runs/ev_<variant>_s<seed>
python -m scripts.train_mappo --phase seeker --config configs/mappo_config_phaseB.json \
    --opponent-model models/ppo_city_plain_s2001.zip --output runs/mappo_B --seeds 2001
# watch: PyBullet 3D window and 2D GIF
python -m scripts.replay_stage2 --joint-checkpoint runs/C/simultaneous_s2001/final.pt --seed 10003
python -m scripts.view2d --matchup joint --obstacles city \
    --joint-checkpoint runs/C/simultaneous_s2001/final.pt --out joint.gif
```

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